

Customer Information

Madam Pearl

Ghana

Accra, Ghana

Quote #: EP-2025-00028**Date:** January 20, 2026**Valid for:** 30 days

OPTION 1

HYBRID SYSTEM — PRICE: GHS 94,884

Panels: 8 Longi Solar panels of 570W.

Inverter: 6.5 kW Hybrid inverter.

Batteries: 10 kWh LiFePO4 Lithium Battery.

MAXIMUM OF 3500 WATTS.

The system will power: 1 standard fridge, 1 blender, 1 sound system, 1 iron, 1 rice cooker,
1 AC (1 or 1.5 HP max), 25 bulbs, 5 fans, 2 TVs, 3 computers/laptops, printer.

Best for clients focused on reducing bills and short-term backup for lights and lighter loads.

Item	Unit price (GHS)	Qty	Amount (GHS)
570W Longi Solar Panel	1,500.00	8.0	12,000.00
PV mid / end clamp (each) (3.5× panels)	30.00	28.0	840.00
MC4 connector pair (~1.2 pairs per panel)	25.00	10.0	250.00
6.5 kW Hybrid Inverter	6,500.00	1.0	6,500.00
RR 10MM Copper Cable (100m coil) (AC feeder coils vs inverter kW)	2,500.00	2.0	5,000.00
AC isolator switch 40A (AC isolator per inverter)	280.00	1.0	280.00
100A changeover (1 per inverter / genset path)	1,500.00	1.0	1,500.00
5 kWh Lithium Ion Battery	13,000.00	2.0	26,000.00
Battery cable 35mm copper (per meter) (~2 m per kWh)	80.00	20.0	1,600.00
250A MCCB battery breaker (battery bank protection)	650.00	2.0	1,300.00
Mounting structure (rails) (proforma rail sets)	350.00	5.0	1,750.00
Transportation of logistics	3,000.00	1.0	3,000.00
Installation & commissioning (standard) (tier for 4.6 kW)	6,000.00	1.0	6,000.00

Item	Unit price (GHS)	Qty	Amount (GHS)
Single core 4mm PV cable (100m coil) (PV homerun (1 DC strings))	2,500.00	2.0	5,000.00
AC cable 6 mm ² (100m coil) (AC branch / DB wiring)	1,800.00	1.0	1,800.00
63A DC MCB 2P (DC string groups (1 strings))	200.00	2.0	400.00
DC isolator switch 32A (DC isolator per string group)	350.00	1.0	350.00
100A AC breaker (AC branch protection)	250.00	1.0	250.00
6-way breaker box	180.00	1.0	180.00
13-way breaker box	350.00	2.0	700.00
SPD / lightning arrester (500V class) (standard hybrid)	300.00	1.0	300.00
Earthing kit (rod, clamp, 16mm earth cable) (earthing per array section)	450.00	2.0	900.00
PVC trunking 25mm (per metre) (trunking metres)	35.00	12.0	420.00
DC/AC labeling & warning kit (labels & warnings)	120.00	1.0	120.00
Miscellaneous (MC4, trunking, PVC, glands, fixings, lugs, ties) (remaining consumables (glands, lugs, ties))	1,500.00	1.0	1,500.00

Sub total:	77,940.00
Tax (20.0%):	15,588.00
Discount (0%):	0.00
Option total (GHS):	93,528.00

OPTION 2

HYBRID SYSTEM — PRICE: GHS 154,644

Option 2 upgrades backup storage to accommodate more appliances.

- Panels: 16 Longi Solar panels of 570W.
- Inverter: 10 kW Hybrid inverter.
- Batteries: 20 kWh LiFePO4 Lithium Battery.

The system will power: 2 standard fridges/freezers, 1 blender, 1 sound system, 1 iron, 1 rice cooker/microwave, 2 ACs (1 / 1.5 / 2.5 HP), 30 bulbs, 4 fans, 2 TVs, 3 computers/laptops/printer.

Item	Unit price (GHS)	Qty	Amount (GHS)
570W Longi Solar Panel	1,500.00	16.0	24,000.00
10 kW Hybrid Inverter	11,000.00	1.0	11,000.00
5 kWh Lithium Ion Battery	13,000.00	4.0	52,000.00
Battery Cable 50mm Copper Cable	200.00	18.0	3,600.00
Battery Cable 35mm Copper Cable	90.00	10.0	900.00
250A MCC Battery Breaker	650.00	2.0	1,300.00
Mounting Structure (Rails)	350.00	8.0	2,800.00
Transportation of Logistics	3,000.00	1.0	3,000.00
Installation & Commissioning	8,000.00	1.0	8,000.00
Mid clamps (32), End clamps (16)	30.00	48.0	1,440.00
RR 10MM AC Cable (100m/coil)	2,500.00	2.0	5,000.00
Single Core 6mm PV Cable (100m/coil)	2,500.00	4.0	10,000.00
63A DC MCB 2P breakers	200.00	3.0	600.00
100A AC Breaker	250.00	1.0	250.00
125A AC Breaker	300.00	1.0	300.00
100A Change Over	1,500.00	1.0	1,500.00
6-way Breaker Box	180.00	1.0	180.00
13-way Breaker Boxes	350.00	2.0	700.00
SPD lightning arrester, 1000V	300.00	1.0	300.00
Miscellaneous (MC4, trunking, PVC, glands, fixings, etc.)	2,000.00	1.0	2,000.00

Sub total:
128,870.00

Tax (20.0%):
25,774.00

Discount (0%):
0.00

Option total (GHS):
154,644.00

SYSTEM SPECIFICATIONS & ENGINEERING DETAILS

DAILY ENERGY REQUIREMENT

15.26 kWh/day

EFFECTIVE DAILY ENERGY

21.19 kWh/day

After system losses

MONTHLY GENERATION

635.7 kWh

PEAK SUN HOURS

5.0 hrs/day

Location: N/A

PV ARRAY CAPACITY

5.13 kW

9 × 570W panels

PANEL BRAND

Longi

570W model

INVERTER CAPACITY

3.9 kW

Selected: 5.0kW
(for quote)

DC/AC RATIO

1.03

Optimized

BATTERY CAPACITY

5.0 kWh

BACKUP DURATION

10 hours

REQUIRED ROOF AREA

28.1 m²

MOUNTING RAILS (EST.)

5 sticks (18 ft)

≈ 22.6 m linear

SYSTEM EFFICIENCY

72.0 %

Engineering Notes: This system is designed using industry-standard engineering factors, including system efficiency (accounting for inverter, wiring, and temperature), an optimized DC/AC ratio to prevent inverter clipping, and location-specific peak sun hours for accurate energy-generation calculations. All calculations include appropriate safety margins for reliable performance.

Mounting rail stick count is a **planning estimate** (18 ft segments, portrait grid); confirm with site layout and structural design.

Terms & Conditions

This quotation is valid for 30 days from the date of issue. All prices are subject to change without notice. Installation and project delivery terms will be discussed upon acceptance.

Payment Terms

30% deposit · 40% on delivery · 30% on completion

Notes

RESIDENTIAL PROFORMA — Madam Pearl Invoice no: EP-2025-00028 · TIN: C0008609403 Document date: 20 January 2026 · Valid 30 days

Thank you for your business!